

Question Paper

Exam Date & Time: 14-Jun-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - JUNE 2025

SUBJECT: PHY 1071/PHY_1071 - ENGINEERING PHYSICS

(MAKE-UP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) With a necessary diagram, derive an expression for angle of acceptance and numerical aperture. (4)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) A three level laser emits light of wavelength 550 nm. (a) What is the ratio of population of the upper level (E_2) to that of the lower level (E_1) in laser transition, at 300 K? (b) At what temperature the ratio of the population of E_2 to that of E_1 becomes half? (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) A ruby laser emits light at a wavelength of 694.4 nm. If a laser pulse is emitted for 12.0 ps and the energy release per pulse is 150 mJ, what is the length of the pulse, and how many photons are there in each pulse? (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Define (i) group speed and (ii) phase speed of a wave packet. Show that the group speed of a wave packet is equal to the particle speed for a free non- relativistic quantum particle. (4)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2B) Molybdenum has a work function of 4.2 eV. (4)
i) Find the cut off wavelength and cut off frequency for the photoelectric effect.
ii) What is the stopping potential if the incident light has wavelength of 180 nm? ($1 \text{ eV} = 1.6 \times 10^{19} \text{ J}$ and $h = 6.63 \times 10^{-34} \text{ J-s}$)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Illustrate the de Broglie concept of wave properties of particles. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Apply the Schrödinger equation to a particle in a one-dimensional "box" of length L and obtain the energy values of the particle. (4)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3B) The $J = 0$ to $J = 1$ rotational transition of the CO molecule occurs at a frequency of $1.15 \times 10^{11} \text{ Hz}$. (4)

Use this information to;

i) Calculate the moment of inertia of the molecule.

ii) Calculate the bond-length of the molecule.

iii) What happens if another photon of frequency 1.15×10^{11} Hz is incident on the CO molecule while it is in the $J = 1$ state?

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) A proton in a one-dimensional box has an energy of 400 keV in its first excited state. How wide is the box? (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) Explain atomic, ionic, orientational and space charge polarizability using suitable diagram. (4)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4B) Consider a semiconductor in the form of a rectangular strip of thickness $10 \mu\text{m}$. Let a current of 1 mA flow along its length and a transverse magnetic field of strength 1T be applied across its thickness. If charge carrier density is $3.1 \times 10^{23} \text{ m}^{-3}$, then calculate Hall coefficient and Hall voltage. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) Explain why ferroelectrics are known as non-linear dielectrics. Sketch the ferroelectric hysteresis loop. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5A) Explain metallic glass. Write any two property and applications of metallic glass. (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5B) Obtain an expression for rotational energy of a diatomic molecule. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5C) Calculate the surface to volume ratio of a spherical body which has radius of 100 nm. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025

PHY 1071-PHY:ENGINEERING PHYSICS

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	16
4	4	CO 4	8
5	5	CO 5	6
Total			50

* **NA** - CO Not Selected

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PHY 1071-PHY:ENGINEERING PHYSICS

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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	20(40%)
3	Application	10	30(60%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)